

Python has a Plethora of Tools for Cyber Security and Digital Forensics



The increasing use of Internet enabled devices has led to a higher incidence of cyber threats, through various forms of malware. Python is used a lot in cyber security and digital forensics. To cite a common use case, today, videos and images go viral very easily and can be security threats. Python tools can not only identify the location of the image, the camera which first took the video but also the exact smartphone used. Criminals beware! Python is watching you.

Nowadays, most people depend on smartphones and technology based devices for routine work, whether official communication or social media. The prolific use of smart devices and Internet connected gadgets invites cyber attacks and the entire world is affected by it. In India too, cyber attacks are very common.

The Annual Threat Report 2019 by Quick Heal states that the metro cities of India – Delhi, Bengaluru, Mumbai and Kolkata – are the most affected by cyber attacks. The reports underlines that there were more than 950 million

attack instances registered last year. Another report from *inc42.com* states that India is the second most affected country, in terms of cyber attacks of different types. The report calculates average cost per cyber attack to be around Rs 4,552. These figures are scary and, hence, awareness about cyber security and digital forensics should be increased.

The Internet of Things (IoT) is also affected by cyber attacks. Millions of smart devices are connected with each other. These include smart watches, webcams, e-health gadgets and many others. The *Economic Times* reports

that there has been 22 per cent growth in cyber attacks on IoT deployments in India. The research analytics shows that there were more than 2500 unique malware impressions that affected the IoT environment and deployments.

An urgent need for awareness about vulnerabilities and security

These figures and research reports are frightening and must be addressed by law enforcement and forensic investigation agencies. The cyber investigation teams and government officials are required to be equipped with advanced tools and knowledge about programming languages so that any malicious attack can be identified and the root cause traced.

There are a number of software libraries, frameworks, tools and programming languages available for cyber security and digital forensics, which every citizen should be aware of. Awareness is the first line of defence against cyber attacks.